1

a) $(100111.11011)_2$

$= (1\times2^5 + 0\times2^4 + 0\times2^3 + 1\times2^2 + 1\times2^1 + 1) + (1\times2^{-1} + 1\times2^{-2} +$

$0\times2^{-3} + 1\times2^{-4} + 1\times2^{-5})$

$= 39 + 0.84375$

$= (39.84375)_{10}$

c) $(79B.2FG)_{17}$

$= (7\times17^2 + 9\times17^1 + 11\times17^0) + (2\times17^{-1} + 15\times17^{-2} + 16\times17^{-3})$

$= 2187 + 0.1728$

$= (2187.1728)_{10}$

2 a) $(107.0425)_{10}$

$$
\begin{array}{r|l}
2 & 107 \\ \hline
2 & 53 - 1 \\ \hline
2 & 26 - 1 \\ \hline
2 & 13 - 0 \\ \hline
2 & 6 - 1 \\ \hline
2 & 3 - 0 \\ \hline
2 & 1 - 1 \\ \hline
 & 0 - 1
\end{array}
$$

$$
\begin{array}{r}
0.0425 \\
\times\ 2 \\ \hline
0.085 \\
\times\ 2 \\ \hline
0.17 \\
\times\ 2 \\ \hline
0.34
\end{array}
$$

$= (1101011.000)_2$

b) $(6045)_7$

$= 6 \times 7^3 + 0 \times 7^2 + 4 \times 7^1 + 5 \times 7^0$

$= (2091)_{10}$

Now,

$$
\begin{array}{r|l}
15 & 2091 \\ \hline
15 & 139 - 6 \\ \hline
15 & 9 - 4 \\ \hline
 & 0 - 9
\end{array}
$$

$\therefore (946)_{15}$

4

(a) Addition

$$\begin{array}{rr} & (4EB4)_{17} \\ (+) & (9B)_{17} \\ \hline & (4F3F)_{17} \end{array}$$

$20 \% 7 = 3$

$$\begin{array}{r} 17 \overline{)20} (1 \\ 17 \\ \hline 3 \end{array}$$

Subtraction

$$\begin{array}{rr} & 4EB4 \\ (-) & 9B \\ \hline & 4E1A \end{array}$$

$$\begin{array}{r} 4+17 = 21 \\ -11 \\ \hline 10 = A \end{array}$$

Multiplication

$$\begin{array}{r} 4EB4 \\ (x)\quad 9B \\ \hline 3284A \\ 29CG2X \\ \hline 2CF76A \end{array}$$

$11 \times 4 = 44$

$44 \% 17 = 10$

$$17\overline{)44}(2 \quad \begin{array}{r} 34 \\ \hline 10 \end{array}$$

$11 \times 11 = 121 + 2 = 123$

$123 \% 17 = 4$

$$17\overline{)123}(7 \quad \begin{array}{r} 119 \\ \hline 4 \end{array}$$

$11 \times 14 = 154 + 7 = 161$

$161 \% 17 = 8$

$$17\overline{)161}(9 \quad \begin{array}{r} 153 \\ \hline 8 \end{array}$$

$11 \times 4 = 44 + 9 = 53$

$53 \% 17 = 2$

$$17\overline{)53}(3 \quad \begin{array}{r} 51 \\ \hline 2 \end{array}$$

$9 \times 4 = 36$

$36 \% 17 = 2$

$$17\overline{)36}(2 \quad \begin{array}{r} 34 \\ \hline 2 \end{array}$$

$9 \times 11 = 99 + 2 = 101$

$101 \% 17 = 16 = G$

$$17\overline{)101}(5 \quad \begin{array}{r} 85 \\ \hline 16 \end{array}$$

$9 \times 14 = 126 + 5 = 131$

$131 \% 17 = 12$

$$17\overline{)131}(7 \quad \begin{array}{r} 119 \\ \hline 12 \end{array}$$

$9 \times 4 = 36 + 7 = 43$

$43 \% 17 = 9$

$$17\overline{)43}(2 \quad \begin{array}{r} 34 \\ \hline 9 \end{array}$$

$24 \% 17 = 7$

$$17\overline{)24}(1 \quad \begin{array}{r} 17 \\ \hline 7 \end{array}$$

5

a) $(4432)_5 \div (13)_5$

$$\begin{array}{r} {}^{1}13 \\ \times 2 \\ \hline 31 \end{array} \qquad \begin{array}{r} {}^{1}13 \\ \times 3 \\ \hline 44 \end{array} \qquad \begin{array}{r} {}^{2}13 \\ \times 4 \\ \hline 112 \end{array} \qquad \begin{array}{r} {}^{3}13 \\ \times 5 \\ \hline 130 \end{array}$$

$13 \times 1 = 13$

$13 \times 2 = 31$

$13 \times 3 = 44$

$13 \times 4 = 112$

$13 \times 5 = 130$

$6 \% 5 = 1$ $\quad 9 \% 5 = 4$ $\quad 12 \% 5 = 2$ $\quad 15 \% 5 = 0$

$8 \% 5 = 3$

$$5\overline{)6}(1 \quad \begin{array}{r} 5 \\ \hline 1 \end{array} \qquad 5\overline{)9}(1 \quad \begin{array}{r} 5 \\ \hline 4 \end{array} \qquad 5\overline{)12}(2 \quad \begin{array}{r} 10 \\ \hline 2 \end{array} \qquad 5\overline{)8}(1 \quad \begin{array}{r} 5 \\ \hline 3 \end{array}$$

That is,

$$13\overline{)4432}(0302$$

$$\begin{array}{r} 0 \\ \hline 44 \\ 44 \\ \hline 32 \\ 31 \\ \hline 1 \end{array}$$

That is $(0302)_5$ ~~so~~ is the result with $(1)_5$ as remainder.

6 $(F1)_{16}$

$= (15 \times 16^1 + 1 \times 16^0)$

$= (241)_{10} \times 2$

$= (482)_{10}$ for 2 RAMs

Now,

$(10111100111)_2$

$= 1 \times 2^{10} + 0 \times 2^9 + 1 \times 2^8 + 1 \times 2^7 + 1 \times 2^6 + 1 \times 2^5 + 0 \times 2^4 +$

$0 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$

$= (1511)_{10}$

Again,

$(3376)_8$

$= 3 \times 8^3 + 3 \times 8^2 + 7 \times 8^1 + 6 \times 8^0$

$= (1790)_{10}$

$\therefore$ Total money needed $= (482 + 1511) = (1993)_{10}$

Extra money needed $= (1993 - 1790)$

$= (203)_{10}.$

So I won't have any ~~left~~ left but I would need

$(203)_{10}$ dollars more

7

a) ~~A~~ To present excess 6 bits, we need -

$$\begin{array}{r} 9 \\ +6 \\ \hline 15 \end{array}$$

8	4	2	1
1	1	1	1

∴ We need 4 bits to present it.

Now,

Decimal	BCD	Excess 6
4	0100	1010
7	0111	1101
2	0010	1000
0	0000	0110
9	1001	1111

That is -

$$(1010\ 1101\ 1000 \cdot 1000\ 0110\ 1111\ 1101)_{\text{Excess } 6}$$

b) $(\underbrace{0101}_{5}\ \underbrace{0011}_{3}\ \underbrace{0110}_{6}\ \underbrace{1100}_{12}\ \underbrace{1010}_{10} \cdot \underbrace{0100}_{4}\ \underbrace{1001}_{9})$ Excess 3

That is

$$\begin{array}{ccccccc} 5 & 3 & 6 & 12 & 10 \cdot & 4 & 9 \\ -3 & -3 & -3 & -3 & -3 & -3 & -3 \\ \hline 2 & 0 & 3 & 9 & 7 & 1 & 6 \end{array}$$

$\therefore$ The number is -

$$(20397 \cdot 16)_{10}$$

~~Now~~

Now,

$$
\begin{array}{r|l}
8 & 20397 \\
8 & 2549 - 5 \\
8 & 318 - 5 \\
8 & 39 - 6 \\
8 & 4 - 7 \\
 & 0 - 4
\end{array}
$$

$$
\begin{array}{r}
.16 \\
\times 8 \\
\hline
1.28 \\
\times 8 \\
\hline
2.24 \\
\times 8 \\
\hline
1.92 \\
\times 8 \\
\hline
7.36
\end{array}
$$

$\therefore$ The final number $= (47655.1217.....)_8$

8

(a) $+(147)_{10}$

$(147)_{10}$ to binary $\rightarrow$

8 4 2 1   8 4 2 1   8 4 2 1

0 0 0 1   0 1 0 0   0 1 1 1

$\therefore$ 1

$$
\begin{array}{r|l}
2 & 147 \\
2 & 73 - 1 \\
2 & 36 - 1 \\
2 & 18 - 0 \\
2 & 9 - 0 \\
2 & 4 - 1 \\
2 & 2 - 0 \\
2 & 1 - 0
\end{array}
$$

$\therefore (10010011)_2$

Representing in 9 bit –

$$(\underbrace{0}_{\text{Sign}}\underbrace{10010011}_{\text{Magnitude}})_2$$

That is, all three sign-magnitude, 1's compliment, 2's compliment representation will be $(010010011)_2$

b) $-(193)_{10}$

Binary –

$$\begin{array}{r|l}
2 & 193 \\ \hline
2 & 96 - 1 \\ \hline
2 & 48 - 0 \\ \hline
2 & 24 - 0 \\ \hline
2 & 12 - 0 \\ \hline
2 & 6 - 0 \\ \hline
2 & 3 - 0 \\ \hline
2 & 1 - 1 \\ \hline
 & 0 - 1
\end{array}$$

$\therefore (11000001)_2$

Now, representing in 9 bit –

$$(\underbrace{0}_{\text{Sign}}\underbrace{11000001}_{\text{Magnitude}})_2$$

That is,

Sign-magnitude form = $(111000001)_2$

1's compliment –

$(100111110)_2$

2's compliment –

$$\begin{array}{r} 100111110 \\ +\ 1 \\ \hline (100111111)_2 \end{array}$$

9 $(-25)_{10} - (97)_{10} = (-25)_{10} + (-97)_{10}$

Here,

$$\begin{array}{l|l} 2 & 25 \\ 2 & 12-1 \\ 2 & 6-0 \\ 2 & 3-0 \\ 2 & 1-1 \\ & 0-1 \end{array}$$

$\therefore (11001)_2$

$$\begin{array}{l|l} 2 & 97 \\ 2 & 48-1 \\ 2 & 24-0 \\ 2 & 12-0 \\ 2 & 6-0 \\ 2 & 3-0 \\ 2 & 1-1 \\ & 0-1 \end{array}$$

$\therefore (1100001)_2$

Representing in 7 bits –

$25_{10} = (0011001)_2$ $\qquad$ $97_{10} = (1100001)_2$

1's compliment –

$(1100110)_2$ $\qquad$ $(0011110)_2$

2's compliment – $\quad + 1$ $\qquad$ $+1$

$(1100111)_2$ $\qquad$ $(0011111)_2$

Here,

$$\begin{array}{r} 1111111 \\ 1100111 \\ (+)\ 0011111 \\ \hline 1\ 0000110 \end{array}$$

$2 \% 2 = 0$

$3 \% 2 = 1$

$-25 - 97 = -122$

In 2's compliment, extra bits are discarded. So the sum will be $\cancel{(1111110)_2}$ $(0000110)_2$.

That is, ~~t~~ expected output $\neq$ calculated output.

$\therefore$ There is overflow.